

Week 07: Shortest Paths and Graph Cycles

1 Exercises

Exercise 1. Consider a graph G defined by the following distance matrix:

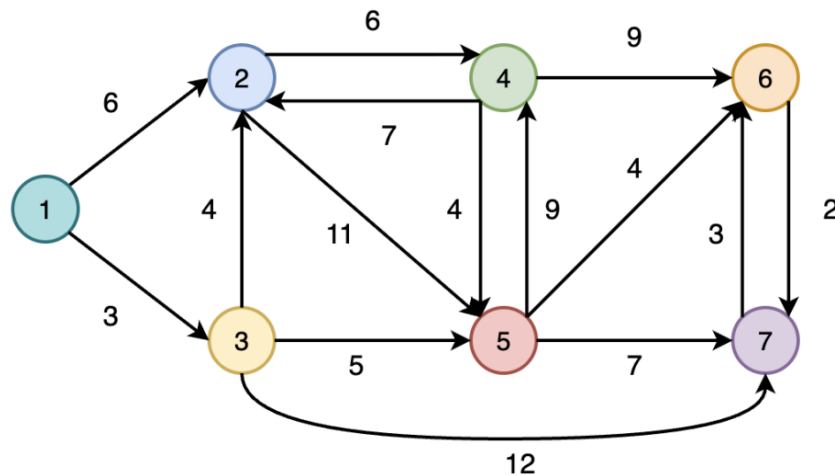
$$G = \begin{pmatrix} 0 & 2 & 5 & \infty & \infty & \infty & \infty \\ \infty & 0 & 1 & 4 & \infty & \infty & \infty \\ \infty & \infty & 0 & 3 & 1 & \infty & \infty \\ \infty & \infty & \infty & 0 & 2 & 6 & \infty \\ \infty & \infty & \infty & 1 & 0 & 3 & \infty \\ \infty & \infty & \infty & \infty & \infty & 0 & 4 \\ \infty & \infty & \infty & \infty & \infty & \infty & 0 \end{pmatrix}$$

Find the shortest path from vertex 1 to vertex 7 given that:

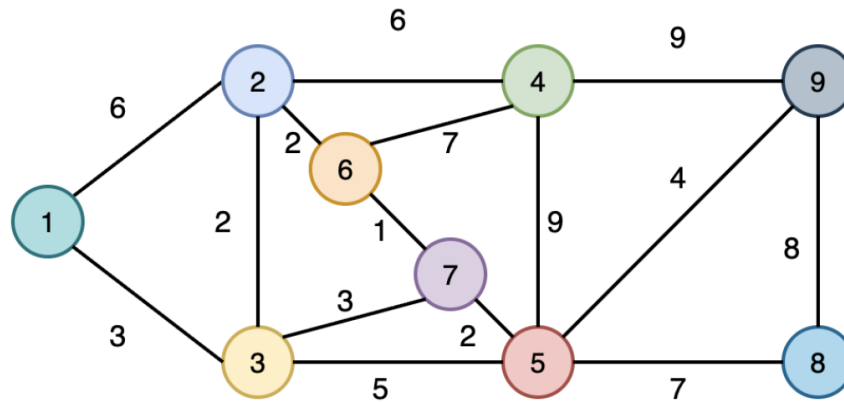
- (a) There are no additional conditions.
- (b) The path does not pass through vertex 5.
- (c) The path must pass through vertex 4.
- (d) The path must pass through the edge $(5, 4)$.

Exercise 2. Find the shortest path from vertex 1 to all other vertices in the graph mentioned above.

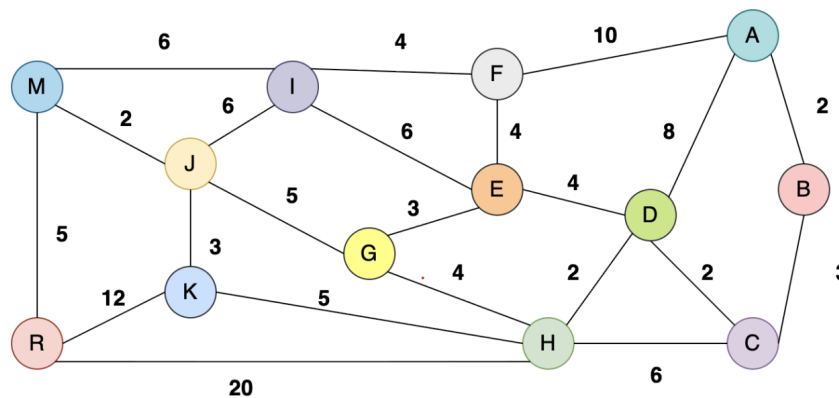
Exercise 3. Find the shortest path from vertex 1 to all other vertices in this graph:



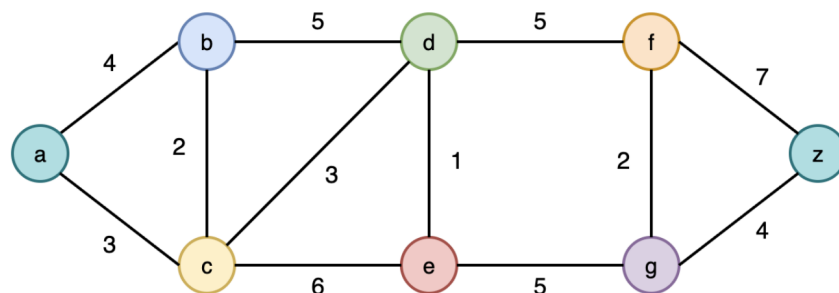
Exercise 4. Find the shortest path from vertex 1 to all other vertices in this graph:



Exercise 5. Find the shortest path from vertex A to vertex R in this graph:



Exercise 6. Find the shortest path from vertex a to vertex z in this graph:



Exercise 7. Is it possible to move a Knight on a 4×4 chessboard such that it visits every square exactly once and returns to the starting square?

(Homework): Think about other cases: for example 5×6 , 8×8 , etc. Hint: For an $m \times n$ board, does the parity of m and n affect the existence of such a path? Consider the size of m and n as well.

Exercise 8. A table tennis tournament with $n \geq 2$ athletes is held in a round-robin format. Prove that when the tournament ends, the athletes can be arranged in a vertical line such that each person in line defeated the person immediately behind them.

Exercise 9. Answer these questions:

- a) Given a complete graph G with 6 vertices numbered from 1 to 6. How many Hamilton paths are there on G starting from vertex 1?
- b) Given a complete graph G with 100 vertices. What is the minimum number of edges that must be removed so that the resulting graph has an Euler circuit?

Exercise 10. Answer these questions:

- a) Prove that a simple, undirected graph where every vertex has a degree of at least 2 contains a simple cycle. Show that if every vertex has a degree exactly equal to 2, it can be partitioned into disjoint simple cycles.
- b) There are $2n$ chess players, each playing exactly 2 matches: one on Day 1 and one on Day 2 (opponents are different). Prove that it is possible to select n players such that no two of them have played each other.